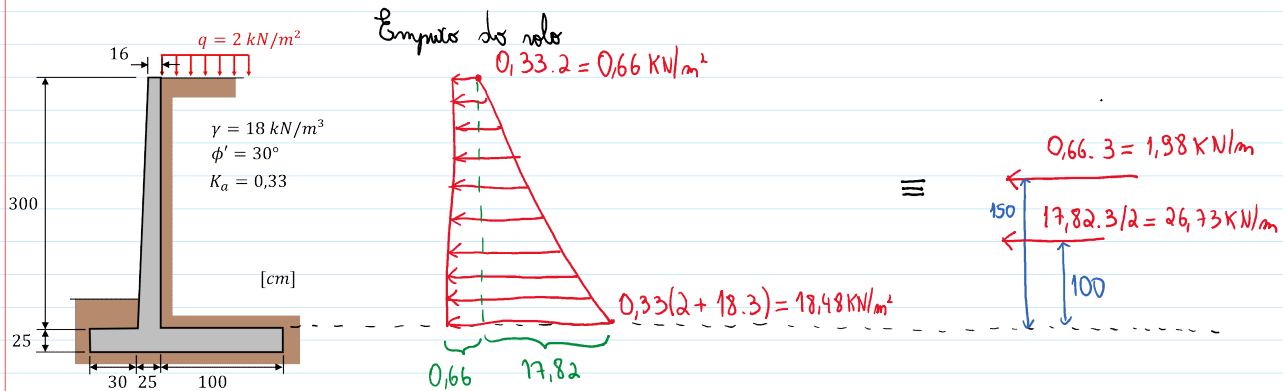


Muro de Arrimo em Balanço



Armadura de flexões

Momentos fletor na base

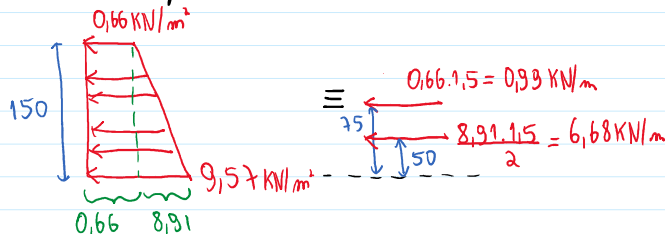
$$M_K = 1,98 \cdot 1,5 + 26,73 \cdot 1 = 29,7 \text{ kNm/m}$$

$$M_d = 1,4 \cdot 29,7 = 41,58 \text{ kNm/m} = 4158 \text{ kNm}$$

$$\left. \begin{array}{l} C25 \\ h = 25 \text{ cm} \\ d = 20 \text{ cm} \end{array} \right\} A_s = 4,96 \text{ cm}^2/\text{m}$$

$$\left. \begin{array}{l} \phi 8 \text{ c}/15 \text{ cm} + \phi 8 \text{ c}/30 \text{ cm} \\ 3,25 \quad 1,68 \end{array} \right\}$$

Momentos fletor no método da altura



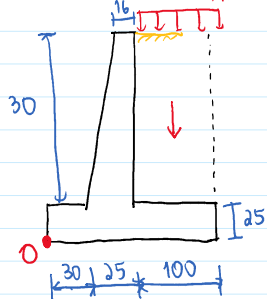
$$M_K = 0,99 \cdot 0,75 + 6,68 \cdot 0,50 = 4,08 \text{ kNm/m}$$

$$M_d = 1,4 \cdot 4,08 = 5,72 \text{ kNm/m} = 572 \text{ kNm}$$

$$\left. \begin{array}{l} h = 20,5 \text{ cm} \\ d = 15,5 \text{ cm} \end{array} \right\} A_s = 3,08 \text{ cm}^2/\text{m}$$

$$\phi 8 \text{ c}/15 \text{ cm}$$

Tombamento



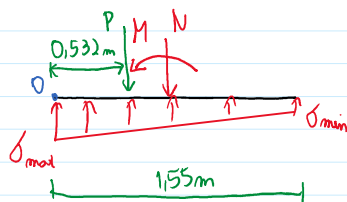
	Peso	Brasos de alavanca	Momentos
Solo:	$18 \cdot 3 \cdot 1 = 54 \text{ kN}$	$0,30 + 0,25 + 0,50 = 1,05 \text{ m}$	$54 \cdot 1,05 = 56,7 \text{ kNm}$
Sobrecarga	$2 \cdot 1 = 2 \text{ kN}$	$1,05 \text{ m}$	$2 \cdot 1,05 = 2,1 \text{ kNm}$
Parade:	$25 \cdot (0,16 + 0,25) \cdot 3 = 15,37 \text{ kN}$	$0,30 + 0,125 = 0,425 \text{ m}$	$15,37 \cdot 0,425 = 6,53 \text{ kNm}$
Fundações:	$25 \cdot 1,55 \cdot 0,25 = 9,69 \text{ kN}$	$1,55/2 = 0,775 \text{ m}$	$9,69 \cdot 0,775 = 7,51 \text{ kNm}$
	$\Sigma 81,06 \text{ kN}$		$\Sigma 72,84 \text{ kNm}$

$$F.S. \text{ tombamento} = \frac{M_{estabilizante}}{M_{instabilizante}} = \frac{72,84}{29,7} = 2,45 > 1,2 \cdot 1,4 = 1,68$$

Deslizamento

$$F.S. \text{ deslizamento} = \frac{P_{total} \cdot \tan \phi'}{H_{at}} = \frac{81,06 \cdot \tan 30^\circ}{28,71} = 1,63 \approx 1,68 \text{ OK!}$$

Tensões na base



$$M_0 = 72,84 - 29,7 = 43,14 \text{ kNm} \quad e_0 = 43,14 / 81,06 = 0,532 \text{ m}$$

$$e = 1,55/2 - 0,532 = 0,243 \text{ m}$$

$$\sigma_{max} = \frac{81,06}{1,55} + \frac{81,06 \cdot 0,243}{1,55^2/6} = 101,5 \text{ kN/m}^2 \sim 10 \text{ tf/m}^2$$

$$\sigma_{min} = \frac{81,06}{1,55} - \frac{81,06 \cdot 0,243}{1,55^2/6} = 3,1 \text{ kN/m}^2$$

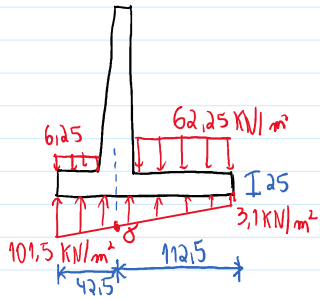
comparar com a carga do solo

Armaduras da fundação

$$\text{Solo: } 3 \cdot 18 = 54 \text{ kN/m}^2$$

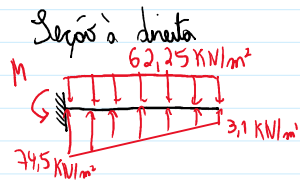
$$\sigma = 3,1 + \frac{(101,5 - 3,1) \cdot 1,125}{1,55} = 74,5 \text{ kN/m}^2$$

momentos de fôrças



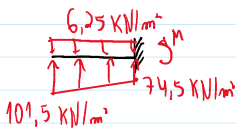
$\text{Lobo: } 3 \cdot 18 = 54 \text{ kN/m}^2$
 $\text{Sobrecarga: } 2 \text{ kN/m}^2$
 $\text{Depota: } 0,25 \cdot 25 = 6,25 \text{ kN/m}^2$
 $\Sigma 62,25 \text{ kN/m}^2$

$\sigma = 3,1 + \frac{101,5 - 3,1 \cdot 1,25}{1,55} = 74,5 \text{ kN/m}^2$



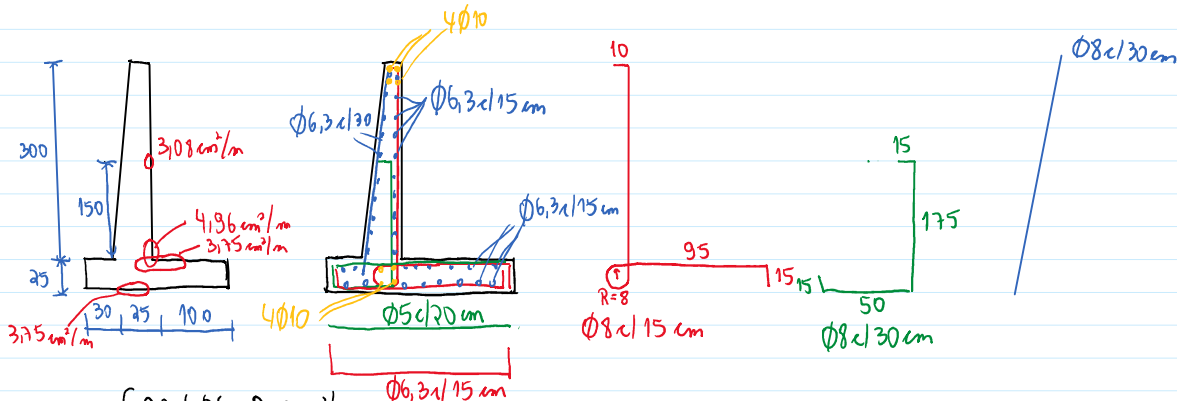
$M_K = 62,25 \cdot \frac{1,25 \cdot 1,25}{2} - 3,1 \cdot \frac{1,25 \cdot 1,25}{2} +$
 $- (74,5 - 3,1) \cdot \frac{1,25 \cdot 1,25}{2} = 22,37 \text{ kN m/m}$
 $M_d = 1,4 \cdot 22,37 = 31,32 \text{ kN m/m} = 3132 \text{ kN cm/m}$
 $A_s = 3,75 \text{ cm}^2/\text{m}$

Seção à esquerda



$M_K = 74,5 \cdot 0,425 \cdot \frac{0,425}{2} + (101,5 - 74,5) \cdot \frac{0,425 \cdot \frac{2}{3} \cdot 0,425}{2} - 6,25 \cdot 0,425 \cdot \frac{0,425}{2} = 7,79 \text{ kN m/m}$
 $M_d = 1,4 \cdot 7,79 = 10,91 \text{ kN m/m} = 1091 \text{ kN cm/m}$
 $A_s = 3,75 \text{ cm}^2/\text{m}$

Detalhamento



$A_{s, \text{dist}} \geq \begin{cases} 0,2 \cdot 4,96 = 0,99 \text{ cm}^2/\text{m} \\ 0,5 \cdot 3,75 = 1,875 \text{ cm}^2/\text{m} \\ 0,9 \text{ cm}^2/\text{m} \end{cases}$